DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111, India

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

INTERNSHIP REPORT
ON

"Pattern analysis of live agent transfers to enhance Virtual Agent self-service"

Submitted in partial fulfillment for the award of the degree of

Bachelor of Engineering

In

Artificial Intelligence and Machine Learning

Submitted by:

Shivabasava Matur- 1DS21AI046

Undertaken at

verizon

Verizon India, Building Number 9, Mindspace, HITEC City, Hyderabad, Telangana 500081

Department of Artificial Intelligence and Machine Learning,

Dayananda Sagar College of Engineering, Bengaluru

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CERTIFICATE

This is to certify that the Internship titled **“Pattern analysis of live agent transfers to enhance Virtual Agent self-service”** carried out by **Shivabasava Matur [1DS21AI46]**, a bonafide student **of Dayananda Sagar College of Engineering**, in partial fulfillment for the award of **Bachelor of Engineering in Artificial Intelligence and Machine Learning** during the academic year **2024–2025**. It is certified that all corrections/suggestions indicated have been incorporated in the report. The internship report has been approved as it satisfies the academic requirements in respect of the course Internship (21INT82).

Signature of the	**Signature of the**	**Signature of the**
Reporting Manager	**Guide**	**Head of the Department**
Rajaraman Narayanaswamy,	**Prof. Kavya D N,**	**Dr. Vindhya P Malagi,**
Manager,	**Assistant Professor,**	**Professor & Head,**
Emerging commercial plt,		
Verizon	**Dept. of AI & ML, DSCE**	**Dept. of AIML, DSCE**

Name of the Examiners	**Signature with date**
1.	**..**
2.	**..**

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DECLARATION

I, **Shivabasava Matur**, a final year student of **Artificial Intelligence and Machine Learning**, **Dayananda Sagar College of Engineering**, **Bengaluru**, hereby declare that this internship report is based on the work undertaken during my ongoing internship in the role of **Intern, Servicenow Devolper** at **Verizon, Hyderabad.**

This report is submitted in partial fulfillment of the requirements for the award of the Bachelor's Degree in Artificial Intelligence and Machine Learning during the academic year 2024–2025.

Date: 12/04/2025 **Place: Bengaluru**

ACKNOWLEDGEMENT

This internship is the result of continuous guidance, support, and encouragement from several individuals, to whom I am deeply grateful. I take this opportunity to express my sincere appreciation to everyone who contributed to the successful completion of this internship.

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Shivabasava Matur [1DS21AI046]

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CHAPTER 1

INTRODUCTION

In the competitive landscape of the retail industry, customer retention and loyalty have emerged as critical factors influencing long-term business sustainability. With rising customer acquisition costs and shifting consumer behaviour, businesses are increasingly turning to data analytics to gain insights into purchasing patterns, customer preferences, and churn risks. Effective customer retention strategies not only enhance revenue growth but also foster stronger brand relationships.

This project focuses on the analysis of customer behaviour using various marketing metrics such as acquisition cohort size, email subscriber engagement, Average Order Value (AOV), and Units per Transaction (UPT). These indicators are instrumental in understanding the effectiveness of marketing campaigns and in identifying high-value customer segments. Analysing these metrics helps in measuring customer lifetime value and in developing personalized marketing interventions.

Cohort analysis forms a core component of this project, offering a structured approach to track the behaviour of customer groups over time based on their acquisition period. This method provides deeper insights into customer engagement trends, retention rates, and the long-term value of different customer segments. By comparing cohorts, the project identifies drop-off patterns and the influence of external factors on customer retention.

To complement the retention analysis, a predictive model for customer churn has been developed. The model utilizes historical transaction data and behavioural attributes to identify customers at risk of disengagement. Key features used include purchase frequency, recency, and order value trends. By applying classification algorithms, the model aims to support proactive decision-making in customer relationship management and reduce churn rates through targeted marketing strategies.

The overall objective of the project is to leverage marketing analytics to drive data-informed decision-making in the retail sector. By combining descriptive and predictive techniques, the project contributes to optimizing retention strategies, enhancing customer loyalty, and improving overall marketing efficiency.

CHAPTER 2
ABOUT THE COMPANY

Verizon India is a wholly owned subsidiary of Verizon Communications Inc., a global leader in telecommunications and technology services. It serves as a vital global delivery center supporting Verizon's operations in areas such as information technology, network engineering, cybersecurity, finance, and business operations. Located primarily in Chennai and Hyderabad, Verizon India plays a crucial role in driving innovation and digital transformation across the enterprise. The teams in India work on cutting-edge technologies like 5G, artificial intelligence, machine learning, cloud computing, and the Internet of Things. Verizon India contributes significantly to software development, application testing, automation, and agile delivery models. It enables seamless collaboration with global counterparts to ensure the efficient rollout of products and services. The company is committed to building a high-performance culture based on continuous learning, employee engagement, and inclusion. With a focus on diversity, Verizon India actively promotes equal opportunities and an inclusive workplace. It provides numerous training programs and learning platforms to help employees upgrade their technical and leadership skills. Employee well-being is a top priority, and the company offers flexible working policies, wellness programs, and mental health support. Verizon India also emphasizes corporate social responsibility through environmental and community-driven initiatives. It contributes to the development of network infrastructure, helping maintain the quality and reliability of Verizon's global communication systems. The company leverages analytics and automation to optimize business processes and enhance customer experiences. It is a key player in improving Verizon's digital presence and transforming customer service with AI-powered tools. Verizon India is also involved in R&D and prototyping of next-generation network solutions. Employees work in a collaborative and tech-savvy environment, fostering innovation and operational excellence. The company recognizes and rewards talent through transparent performance evaluation systems and growth opportunities. It maintains strong ethical values and corporate governance practices. As a trusted partner in Verizon's global strategy, Verizon India ensures resilience, scalability, and technological leadership. The organization's work not only supports millions of customers worldwide but also defines the future of digital connectivity.

CHAPTER 3
SCOPE OF THE WORK

The scope of this project centered around analyzing and optimizing the AYS (AtYourService) Virtual Agent used internally by Verizon India employees through the ServiceNow platform. The primary goal was to reduce the number of interactions that were being escalated to Live Agents, which accounted for 30% of total interactions in 2024. The project aimed to enhance the Virtual Agent's ability to understand user queries, provide accurate responses, and deliver a seamless self-service experience.

The work covered a variety of focus areas, starting with the nalysis of interaction records and Live Agent transcripts. With a total of 9,152 records analyzed, the team sought to understand why escalations were happening and to identify recurring patterns. This required an in-depth review of internal conversation data, which was both time-consuming and complex due to the volume and diversity of queries. Particular attention was paid to knowledge gaps, confusing language, mapping errors, and behavioral trends indicating a preference for human interaction.

Based on the findings, the scope extended to designing and implementing actionable solutions. This included updating 196 utterances, creating 5 new conversation topics, and expanding and restructuring the Knowledge Base (KB). The goal was to ensure that the Virtual Agent could not only interpret a wider variety of intents but also map them effectively to the most relevant KB articles. Additional focus was placed on using ServiceNow's NLU Workbench and Predictive Intelligence to identify gaps and continuously refine the utterance-intent-topic relationships.

Furthermore, the project involved leveraging Virtual Agent Designer and Conversational Interfaces (CI) to rework existing workflows and visualize user interactions more effectively. Although the team didn't operate within a formal scrum framework, the work was collaboratively divided among a team of five, each member contributing to development, testing, and refinement efforts.

Lastly, the scope included recommending future enhancements, such as implementing trigger-based automation that captures resolved live agent interactions and uses them to enrich the virtual agent’s knowledge base. These future-oriented enhancements fall under process improvement and knowledge automation—critical for sustaining the long-term efficiency of the system.

In summary, the scope of work spanned from detailed interaction analysis to practical solution design and deployment within the ServiceNow environment. It demonstrated a comprehensive approach to problem-solving and automation within enterprise customer service operations, laying the groundwork for continued enhancements to the AYS Virtual Agent experience.

CHAPTER 4

LEARNING OBJECTIVES

1. To understand and analyze live agent escalation patterns in a virtual support environment.

One of the key objectives was to gain a thorough understanding of why certain Virtual Agent interactions escalate to live agents. This included analyzing over 6,800 records to identify common issues such as intent mismatches, incomplete workflows, and knowledge gaps. By evaluating escalation triggers, the learning focused on how intelligent automation can improve customer self-service and reduce support workload. This objective helped build a strong foundation in escalation analysis and customer experience optimization using ServiceNow tools.

2. To apply Predictive Intelligence and NLU capabilities for intent mapping and process refinement.

A critical goal was to explore ServiceNow's Predictive Intelligence and NLU Workbench to enhance the accuracy of topic recognition and intent classification. The process involved updating utterances, mapping intents more precisely, and ensuring alignment with user queries. Hands-on work with NLU models and utterance optimization deepened the understanding of conversational AI behavior and how machine learning enhances virtual agent efficiency. This objective reinforced practical application of AI/ML within ITSM environments.

3. To design automated workflows using Virtual Agent Designer to reduce escalation rates.

Another learning objective focused on workflow design using the Virtual Agent Designer. By identifying repetitive escalation categories and converting them into guided automated flows, the goal was to proactively address issues before escalation. This included decision-tree creation, scenario mapping, and testing chatbot responses in real-time. This objective enabled a deeper appreciation of user journey mapping, conversational design, and process automation best practices.

4. To interpret chat transcripts and generate business insights for reducing support costs.

An essential goal was to analyze historical live agent transcripts to uncover communication gaps and missed automation opportunities. This involved identifying keywords, customer frustration points, and unsupported queries. From these insights, business recommendations were formulated to improve topic coverage and deflection strategies. This learning reinforced the connection between data interpretation and operational improvements, highlighting how analytics can directly influence support cost reduction and CX enhancement.

5. To develop reporting and visualization skills for stakeholder communication.

Finally, the project emphasized the importance of data storytelling and reporting. Using performance metrics such as escalation rate reduction, estimated revenue savings, and time saved, dashboards and summary reports were prepared for stakeholder review. Tools like ServiceNow’s performance analytics and external visualization platforms were used to represent findings clearly. This objective sharpened the ability to create actionable visual narratives that translate technical findings into business impact.

CHAPTER 5
CHALLENGES AND SOLUTIONS

1. Dealing with Incomplete or Noisy Data.

A significant challenge in the early stages of the project was managing the quality of customer data, which included missing values, inconsistencies, and noise in key variables such as transaction dates, order values, and email engagement indicators. These issues impacted the reliability of retention and cohort analyses, potentially leading to inaccurate conclusions. To mitigate this, rigorous data cleaning procedures were implemented, including the use of imputation methods to handle missing values, outlier detection to identify anomalies, and filtering techniques to remove irrelevant or duplicate records. This process was critical to ensure data integrity before proceeding with analysis or model development.

2. Identifying Relevant Features for Churn Modeling.

Developing an effective churn prediction model required careful selection of features that were both meaningful and predictive. Given the extensive set of available variables—ranging from transactional metrics to engagement history—feature selection posed a substantial challenge. This was addressed through a structured exploratory data analysis (EDA), where correlation heatmaps, feature importance scoring from baseline machine learning models, and domain expertise were used to identify and prioritize relevant features. The aim was to strike a balance between model interpretability and predictive performance, focusing on features that directly captured customer behavior and purchasing patterns.

3. Balancing Imbalanced Classes in Churn Dataset.

The churn prediction problem was inherently imbalanced, with a majority of customers being retained and a minority labeled as churned. This class imbalance created a challenge for model training, as standard algorithms tended to bias towards the dominant class, reducing the model's ability to correctly identify churned customers. To address this, resampling techniques such as Synthetic Minority Over-sampling Technique (SMOTE) were employed to increase the representation of the minority class. Stratified sampling was also applied during train-test splits to preserve class distribution. Additionally, evaluation metrics such as F1-score, Precision, Recall, and ROC-AUC were emphasized over accuracy to ensure a more balanced and meaningful assessment of model performance.

4. Translating Technical Insights into Business Language.

An ongoing challenge throughout the project was ensuring that complex analytical results could be communicated effectively to non-technical stakeholders. Since the audience for the findings included marketing and business decision-makers, it was essential to translate technical insights into actionable business terms. This was achieved by designing interactive visual dashboards, summary-level data tables, and narrative reports that highlighted key takeaways in an intuitive manner. Emphasis was placed on using clear visual storytelling techniques to bridge the gap between analytics and strategic decision-making, ensuring that insights could drive real-world business outcomes.

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

1. Successful Implementation of Cohort-Based Retention Analysis

One of the key achievements during the internship was the successful application of cohort analysis to segment customers based on acquisition time and observe their behavioral trends over months. The analysis provided critical insights into retention drop-off points and highlighted the effectiveness of previous marketing strategies. This contribution enabled the business to make informed decisions regarding customer lifecycle management and design data-driven engagement campaigns.

2. Development of a Churn Prediction Model

A significant contribution was the design and development of a machine learning model to predict customer churn. The model was trained using relevant customer behavior features such as transaction frequency, average order value, and recency of last purchase. The output provided a churn probability score, allowing the marketing team to identify high-risk customers and take proactive steps to re-engage them. This model is a valuable asset for customer retention efforts.

3. Data Cleaning and Feature Engineering for Business Readiness

An important aspect of the project involved extensive data preprocessing and feature engineering to prepare the dataset for meaningful analysis and modeling. This included handling missing values, removing duplicates, and creating new derived features like customer tenure and average basket size. The cleaned and transformed dataset not only improved model accuracy but also ensured that the insights generated were business-ready and actionable.

4. Insightful Marketing Reports and Visual Dashboards

Another major contribution was the creation of marketing insight reports and interactive dashboards that presented analytical findings in a visually compelling manner. These reports included key metrics on customer retention, cohort performance, and churn predictions, making it easier for marketing teams to track performance and identify areas for improvement. The visualization work bridged the gap between data science and business application.

5. Practical Exposure to Real-World Retail Data and Strategy

The internship provided hands-on experience with real-world data from the retail sector, allowing for an in-depth understanding of customer behavior, sales trends, and marketing effectiveness. Contributing to strategic decision-making through data-backed insights gave a real sense of how analytics directly supports business goals. This practical exposure helped build both technical confidence and business acumen.

6. Enhanced Understanding of Customer Lifecycle Management

Through continuous analysis of customer behavior over various stages — acquisition, engagement, retention, and churn — a deeper understanding of the customer lifecycle in the retail industry was developed. This learning was translated into actionable insights that contributed to designing more targeted and timely marketing strategies. By aligning the business goals with customer journey data, the contribution strengthened the customer-centric approach of the organization.

7. Collaboration with Cross-Functional Teams for Insight Alignment

Effective collaboration with marketing, data, and business intelligence teams played a vital role in aligning analytical findings with strategic objectives. Contributions included participating in discussions, understanding business needs, and adjusting analysis techniques based on team feedback. This cross-functional collaboration ensured that the analytics work was relevant, timely, and impactful, further enhancing the overall value delivered to the organization.

CHAPTER 7
REFLECTION AND ANALYSIS

During my internship at Verizon, I had the opportunity to make significant contributions that advanced the efficiency and scalability of the AYS (Ask Your Solution) Al-powered customer support platform. One of my primary achievements involved the detailed analysis of 6,864 historical customer support records, which provided deep insights into recurring issues that frequently resulted in escalations to live agents. Through this data-driven approach, I was able to identify key pain points such as ambiguous user queries, credential access difficulties, and MAC system configuration errors. Leveraging this understanding, I focused on enhancing the Natural Language Understanding (NLU) model within ServiceNow by meticulously refining the utterance libraries. This involved the addition of over 30 new and updated utterance-intent pairs that better reflected real user behavior, thus significantly improving the system's ability to interpret and respond to a broader range of user inputs.

In parallel, I collaborated with the knowledge management team to address a major gap in the system's ability to resolve queries independently-namely, the absence of adequate and up-to-date Knowledge Base (KB) articles. To bridge this gap, I designed and implemented a workflow that automated the identification of resolved live agent cases that could be converted into structured, reusable knowledge content. This initiative led to the creation and validation of more than 12 high-quality, scenario-based KB articles, which were integrated into the virtual assistant's knowledge loop. These articles directly empowered the virtual agent to provide immediate, accurate responses to common issues without human intervention.

Another major accomplishment was my redesign of the conversation flows using ServiceNow's Virtual Agent Designer. I utilized real conversation logs and user interaction data to create more intuitive and human-like dialog paths, minimizing conversational dead-ends and enhancing user satisfaction. This redesign addressed many instances where users previously experienced frustration due to vague, repetitive, or overly rigid system responses. By incorporating fallback strategies, contextual awareness, and branching logic, the virtual assistant became more resilient and adaptive in handling ambiguous or complex queries.

As a result of these combined efforts, the overall system performance showed measurable improvement. The rate of successful intent recognition increased to 60%, while live agent escalations were reduced by approximately 15%. These outcomes translated into tangible business value, with an estimated operational cost savings of $345,000 and a reduction of over 4,000 agent work hours. Beyond immediate metrics, I also contributed to the platform's long-term success by developing reusable design templates, documenting implementation procedures, and establishing scalable improvement frameworks. These artifacts now serve as reference models for future enhancements and onboarding processes.

Furthermore, my work involved close collaboration with cross-functional teams including UX designers, data analysts, and developers. This coordination ensured that all changes were aligned with Verizon's strategic vision for customer service innovation. My contributions were not only technical but also strategic, reinforcing the importance of user-centered design, continuous feedback integration, and scalable Al development. Overall, my internship at Verizon allowed me to make a lasting impact on the AYS platform while gaining invaluable hands-on experience in enterprise-level Al solutions.

CHAPTER 8
RECOMMENDATIONS

To strengthen customer retention initiatives, it is recommended that the company continue to invest in personalized marketing strategies. By leveraging customer segmentation based on behavioral and transactional data, targeted promotions and tailored communication can be deployed to engage specific customer cohorts more effectively. This approach not only enhances the relevance of marketing efforts but also fosters stronger customer relationships and long-term brand loyalty.

Implementing a real-time churn monitoring system is also advisable. Integrating predictive churn models into the Customer Relationship Management (CRM) platform would enable proactive identification of high-risk customers. Timely interventions, such as personalized outreach or exclusive incentives, can significantly reduce churn and increase customer lifetime value.

In addition, automating and standardizing cohort analysis as part of the regular reporting process would provide ongoing visibility into customer retention patterns. This would allow the business to evaluate the effectiveness of acquisition channels and marketing campaigns over time, enabling data-driven budget allocation and strategy refinement.

Another key recommendation is to enhance data literacy across cross-functional teams. Providing training on fundamental analytics concepts and tools will empower marketing, sales, and product teams to better interpret insights and align their initiatives with data-driven goals. Cultivating a data-literate culture ensures more cohesive collaboration and informed decision-making organization-wide.

Finally, it is essential to maintain a clean, integrated, and scalable data infrastructure. As the volume and complexity of data sources grow, the adoption of a centralized data warehouse or modern analytics platform will help streamline data access, support advanced analytics, and ensure consistency in reporting. This foundational investment will enhance the organization’s agility and analytical maturity in the long term.

CHAPTER 9
CONCLUSION

The internship experience provided a comprehensive understanding of the critical role data-driven decision-making plays in shaping marketing strategies within the retail industry. By working with real-world customer data, the project explored key areas such as customer retention, cohort-based segmentation, and churn prediction, uncovering valuable insights into consumer behavior and business performance. These experiences underscored the strategic impact of marketing analytics in driving informed, outcome-focused decisions.

Hands-on involvement in data preprocessing, feature engineering, predictive modeling, and visualization contributed to the development of strong technical competencies. Simultaneously, translating analytical findings into clear, business-relevant narratives enhanced communication skills. This dual proficiency—technical depth combined with business acumen—proved essential in transforming complex data into actionable insights.

The project also presented various challenges, including managing data quality issues and addressing class imbalance in churn prediction models. Overcoming these obstacles cultivated a problem-solving mindset and built confidence in handling real-world analytical scenarios. Support from mentors and collaborative teamwork played a pivotal role in shaping both technical output and strategic perspective.

Overall, the internship has been a pivotal step toward a career in analytics. It provided a platform to apply academic learning in a professional setting, deepened understanding of business applications of data, and emphasized the importance of aligning analytics with organizational goals. The insights and skills gained during this journey have laid a strong foundation for future growth and success in the field of data and business analytics.